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**RESOURCEHUB: A SECURE CAMPUS RESOURCE SHARING AND EXCHANGE
PLATFORM WITH QR-BASED HANDOVER VERIFICATION**

A CAPSTONE PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Campus Resource Exchange Network(CREN): An Intelligent Academic Marketplace for Resource Sharing**” has been carried out by **Y. Abhay (192411219)**, under the supervision of **DR. D. Sudhagar** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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We, Y. Abhay (192411219) of the CSE Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **“Campus Resource Exchange Network(CREN): An Intelligent Academic Marketplace for Resource Sharing”** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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ABSTRACT

ResourceHub is a student-focused academic resource exchange platform designed to provide a secure and reliable environment for students to exchange books, electronics, laboratory equipment, project components, stationery, and other academic resources. The system combines user authentication, resource management, exchange requests, transaction management, QR-based handover verification, and a peer-based trust system. This report focuses on Module 4: QR Verification & Trust System, which provides secure physical handover verification and establishes a reputation mechanism for students participating in exchanges.

The QR verification component provides an additional security layer between an accepted exchange request and its final completion. After an exchange request is accepted and the resource is reserved, only the resource owner is authorized to generate a temporary QR verification token. The generated token is represented as a QR code and displayed to the owner with an expiry countdown. The requester scans the QR code using the application camera interface or enters the verification code manually when required for testing. The backend validates the token, requester identity, transaction association, and expiration status before marking the handover as verified.

The QR verification mechanism also prevents unauthorized or repeated use of verification tokens. Resource owners cannot verify their own generated QR codes, unrelated students cannot scan tokens belonging to another transaction, expired tokens are rejected and marked as expired, and already verified tokens cannot be reused. QR verification is enforced at the backend before transaction completion, ensuring that an accepted exchange cannot directly move to the completed state without physical handover confirmation.

The trust system complements QR verification by allowing participants in a completed exchange to provide peer ratings and reviews. A rating between one and five stars can be submitted after successful completion of a transaction. The system prevents duplicate reviews for the same transaction and reviewer, validates rating values, and updates the reviewed student's trust score using the implemented rating mechanism. The frontend provides an interactive review interface, rating selection, review text input, submission feedback, and persistent review status after page refresh.

Overall, Module 4 improves the reliability and accountability of ResourceHub by combining physical handover verification with a peer-based reputation mechanism. QR verification confirms that an exchange has reached the intended handover stage, while reviews and trust scores provide students with useful information about the reliability of other participants. The combination of authorization, temporary tokens, expiry validation, transaction protection, review validation, and trust-score calculation provides an additional layer of security and trust to the academic resource exchange platform.

Keywords: QR Verification, Handover Verification, Exchange Transaction, Trust Score, Peer Reviews, Ratings, Secure Token, Resource Exchange, Authorization, ResourceHub.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description	Unit
AES	Advanced Encryption Standard	—
RSA	Rivest–Shamir–Adleman	—
SHA-256	Secure Hash Algorithm – 256-bit	—
GCM	Galois/Counter Mode	—
RSA-OAEP	RSA Optimal Asymmetric Encryption Padding	—
IV	Initialization Vector	Bytes
DEK	Data Encryption Key	Bits
JWT	JSON Web Token	—
API	Application Programming Interface	—
AI	Artificial Intelligence	—
ML	Machine Learning	—
DLP	Data Loss Prevention	—
QR	Quick Response Code	—
URL	Uniform Resource Locator	—
UI	User Interface	—
DB	Database	—
ID	Identifier	—
HTTP	Hypertext Transfer Protocol	—

CHAPTER 1

INTRODUCTION

1.1 Background Information

Students frequently exchange academic resources such as textbooks, laboratory equipment, electronic components, project materials, and stationery. Traditional exchange processes are often handled through informal communication, making it difficult to verify whether an exchange has actually taken place and whether the participating users are trustworthy.

A resource exchange platform can simplify the process by allowing students to list resources, request available resources, and manage exchange transactions through a centralized system. However, simply accepting an exchange request does not guarantee that the physical handover has occurred. A system that directly changes an accepted transaction to completed may incorrectly record an exchange even when the participants have not met or transferred the resource.

ResourceHub addresses this problem through QR-based handover verification. Once an exchange request is accepted and the resource becomes reserved, the resource owner generates a temporary QR verification code. The requester scans the QR code during the physical handover. The backend validates the verification token, confirms that the requester belongs to the transaction, checks its expiration status, and records successful verification before allowing the transaction to be completed.

Another important challenge in peer-to-peer resource exchange is trust. Students need information about the reliability of other users before requesting or accepting resources. ResourceHub therefore introduces a review and trust-score mechanism. After a transaction is completed, participants can rate each other and provide optional written feedback. The resulting review information contributes to the user's trust score and can be displayed to students when evaluating a resource owner or exchange participant.

Module 4, QR Verification & Trust System, therefore provides two important capabilities within ResourceHub. The first capability validates the physical handover of resources, while the second provides a reputation mechanism based on completed exchanges and peer feedback.

1.2 Project Objectives

The main objective of Module 4 is to provide secure handover verification and a reliable trust mechanism for ResourceHub exchanges.

The specific objectives are:

- To generate secure QR verification tokens for accepted exchange transactions.
- To restrict QR generation to authorized resource owners.
- To associate each verification token with the correct exchange transaction.
- To provide temporary QR verification with configurable expiration.

- To allow only the authorized requester to verify the generated QR code.
- To prevent resource owners from verifying their own handover QR codes.
- To reject QR verification attempts from unrelated students.
- To detect and reject expired QR verification tokens.
- To prevent previously verified QR tokens from being reused.
- To ensure that QR verification is completed before an exchange can be marked as completed.
- To maintain transaction consistency during QR verification and exchange completion.
- To allow participants to submit ratings after completed exchanges.
- To support one-to-five-star peer ratings and optional written reviews.
- To prevent duplicate reviews for the same transaction and reviewer.
- To validate review ratings before storing them.
- To calculate and update the user's trust score based on reviews.
- To display trust score and review information to students.
- To provide a persistent frontend review state after page refresh.
- To improve accountability and trust within the ResourceHub platform.

1.3 Significance

The QR Verification & Trust System is significant because it addresses two major challenges in peer-to-peer academic resource exchange: verifying physical handover and establishing user trust.

QR handover verification provides a controlled transition between an accepted transaction and its completion. Instead of assuming that an accepted request has resulted in a successful exchange, ResourceHub requires the participants to complete a verification step during the physical handover. This reduces the possibility of incorrectly completed transactions.

Temporary verification tokens further improve security because a QR code is not valid indefinitely. The system checks the token's expiration time and changes expired verification records to an EXPIRED state. The token is also linked to a specific transaction and requester, preventing unrelated users from using it.

The trust system is significant because peer-to-peer marketplaces depend heavily on participant reliability. A student who has successfully completed several exchanges and received positive reviews can establish a stronger reputation. Conversely, consistently poor ratings can reduce the trust score and provide useful information to other students.

The combination of QR verification and trust scoring strengthens ResourceHub beyond a basic resource listing and request system. It introduces both transaction-level verification and user-level reputation, improving the reliability, accountability, and confidence of the overall platform.

1.4 Scope

The scope of Module 4 covers QR-based physical handover verification, exchange completion protection, peer reviews, ratings, and trust-score management within ResourceHub.

The module includes:

- QR verification token generation.
- Authorization of QR generation by resource owners.
- QR token association with exchange transactions.
- Configurable QR expiration.
- QR code display on the frontend.
- QR scanning using the device camera.
- Manual verification-code input for testing and fallback usage.
- Requester authorization during verification.
- Prevention of owner self-verification.
- Prevention of unrelated-user verification.
- Expired token detection.
- Token reuse prevention.
- Transaction-level QR verification.
- Backend enforcement before transaction completion.
- Database transaction protection during verification.
- One-to-five-star ratings.
- Optional written reviews.
- Duplicate review prevention.
- Review validation.
- Trust-score calculation and updating.
- Review and rating retrieval.
- Trust-score display in the user interface.
- Persistent review status after page refresh.
- End-to-end testing of QR and review functionality.

The module does not replace the authentication, resource management, or exchange request components of ResourceHub. Instead, it works together with these modules to provide additional verification and trust capabilities.

1.5 Methodology Overview

The implementation of Module 4 followed a structured software development approach consisting of requirement identification, database analysis, backend implementation, frontend integration, security validation, and end-to-end testing.

The QR verification process begins when an exchange request has been accepted and the associated resource has been reserved. The resource owner requests generation of a verification token. A secure random token is created and stored with the transaction together with its expiration timestamp. The token is represented as a QR code in the frontend.

During the physical handover, the requester scans the QR code. The backend verifies the transaction association, requester identity, token status, and expiration time. If all conditions are satisfied, the QR verification record is marked as VERIFIED. The transaction can then proceed to completion.

The trust process begins after a transaction reaches the COMPLETED state. Participants can submit a one-to-five-star rating and an optional review. The backend validates the transaction and reviewer authorization before creating the review. Duplicate reviews are rejected and the reviewed user's trust score is updated.

The overall methodology can be represented as follows:

QR Verification Workflow:

Accepted Request → Resource Reserved → Owner Generates QR → Temporary Token → Requester Scans QR → Token Validation → Authorization Check → Expiry Check → QR VERIFIED → Transaction COMPLETED

Trust System Workflow:

Completed Exchange → Review Eligibility Check → Rating & Review Submission → Validation → Store Review → Calculate Trust Score → Display Reputation

Combined Workflow:

Exchange Request → Accepted → Reserved → QR Handover Verification → Completed → Peer Review → Trust Score Update

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Peer-to-peer resource exchange systems face challenges that are not present in simple online purchasing or listing applications. An exchange request being accepted does not necessarily mean that the physical resource has been handed over successfully.

If a system allows an accepted transaction to be marked as completed without additional verification, incorrect transaction states may occur. A user could potentially claim that a handover has occurred even when the physical exchange has not taken place.

Another problem is unauthorized verification. If a verification mechanism uses a simple transaction identifier without a secure token or authorization check, unrelated users may attempt to verify another student's transaction.

Token lifetime is also important. A verification code that remains valid permanently increases the possibility of reuse after the intended handover. Similarly, allowing a verified QR token to be reused could result in multiple verification events for the same transaction.

Trust is another major challenge. Students may not have sufficient information about another participant before requesting a resource. A resource listing alone cannot indicate whether the owner has previously completed exchanges successfully or whether other students consider the owner reliable.

Therefore, ResourceHub requires a mechanism that can:

- Verify physical resource handover.
- Restrict QR generation to authorized users.
- Restrict QR verification to the correct requester.
- Prevent self-verification.
- Prevent unrelated-user verification.
- Expire temporary verification tokens.
- Prevent QR token reuse.
- Protect transaction state changes.
- Require QR verification before completion.
- Allow participants to rate completed exchanges.
- Prevent duplicate or invalid reviews.
- Calculate and display a meaningful trust score.

Module 4 was developed to address these requirements.

2.2 Evidence of the Problem

The need for QR handover verification can be demonstrated through the lifecycle of an exchange transaction. After a request is accepted, the resource becomes reserved, but the system still needs evidence that the participants have physically completed the handover.

Without an additional verification step, the transaction could be changed directly from ACCEPTED to COMPLETED. ResourceHub prevents this by requiring a successful QR verification before completion.

Security testing also demonstrates the need for authorization controls. During implementation, the system successfully blocked QR generation attempts by requesters, blocked attempts by resource owners to scan their own QR codes, and rejected verification attempts by unrelated students.

Token expiration is another important requirement. The implemented system checks the expiration timestamp and changes expired QR verification records to EXPIRED. Attempts to use expired tokens are rejected.

The trust component addresses a separate problem. A student may complete exchanges successfully but still have no visible reputation unless feedback is collected. ResourceHub therefore allows completed exchanges to generate reviews and ratings, which are used to update the participant's trust score.

2.3 Stakeholders

The major stakeholders associated with Module 4 include:

Students / Resource Owners

Resource owners generate QR verification codes for accepted exchanges and benefit from a controlled mechanism for confirming physical handovers.

Students / Requesters

Requesters scan the owner's QR code during the physical handover. They can also provide ratings and reviews after completing an exchange.

Exchange Participants

Both participants contribute to the trust system through peer reviews. Their completed exchange history and received ratings contribute to their reputation.

System Administrators

Administrators can benefit from reliable transaction states and trust information when monitoring the overall platform.

ResourceHub Developers

Developers are responsible for implementing secure token generation, authorization, transaction protection, review validation, and frontend integration.

Academic Institution

The institution benefits from a more structured and accountable environment for students to exchange academic resources.

2.4 Supporting Data/Research

Temporary tokens are commonly used in secure verification systems because they provide an unpredictable value that can be associated with a specific operation. In ResourceHub, cryptographically secure random token generation is used to create QR verification values.

Expiration is another important security principle. A temporary verification mechanism should not remain valid indefinitely because the usefulness and security of the token are limited to a particular handover event. ResourceHub therefore associates each QR verification record with an expiration timestamp.

Authorization is also essential because authentication alone does not determine whether a user is allowed to perform a particular operation. ResourceHub verifies that the QR generator is the resource owner and that the scanner is the requester associated with the transaction.

Database transactions and row locking are used when verification changes important transaction state. These mechanisms help prevent inconsistent updates when multiple requests attempt to modify the same transaction.

Peer ratings are widely used in reputation-based platforms to provide users with information about the reliability of other participants. ResourceHub applies this principle to academic resource exchange by collecting ratings after completed transactions and maintaining a trust score for users.

These principles provide the technical foundation for the QR verification and trust mechanisms implemented in ResourceHub.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

The development of Module 4 followed several stages.

Requirement Identification

The first stage involved identifying the need for a secure handover confirmation mechanism and a reputation system for ResourceHub.

QR Verification Design

A transaction-specific temporary verification mechanism was designed. QR generation was restricted to resource owners after successful request acceptance.

Secure Token Implementation

A cryptographically secure random token was generated and associated with the exchange transaction. The token was stored in the QR verification table along with its status and expiration timestamp.

Authorization Implementation

Backend authorization checks were implemented to ensure that only the resource owner could generate a QR code and only the requester associated with the transaction could verify it.

Expiration and Reuse Protection

QR tokens were designed to expire after a configurable period. Previously verified or expired tokens could not be reused.

Transaction Integration

QR verification was integrated into the exchange completion process. A transaction could not be completed unless a corresponding QR verification record had reached the VERIFIED state.

Review and Rating Implementation

After successful transaction completion, participants were allowed to submit ratings between one and five stars with optional review text.

Trust Score Integration

The review system was integrated with the user's trust score. New reviews update the reviewed participant's reputation according to the implemented trust-score calculation.

Frontend Integration

Interactive QR generation, QR scanning, review submission, rating selection, validation messages, and persistent review states were integrated into the ResourceHub frontend.

Testing and Validation

End-to-end tests were performed to validate authorization, expiry, token reuse prevention, transaction completion, duplicate reviews, invalid ratings, trust-score updates, database rollback, and frontend compilation.

3.2 Tools & Technologies Used

Technology	Purpose
Node.js	Backend runtime environment
Express.js	REST API and backend routing
MySQL	Storage of users, resources, transactions, QR records, and reviews
mysql2	MySQL database connectivity
JWT	Authentication and user identity verification
React	Frontend user interface
Vite	Frontend development and production build
Axios	Frontend-to-backend API communication
qrcode.react	QR code generation in the frontend
html5-qrcode	Camera-based QR code scanning
Crypto module	Secure QR verification token generation
REST API	Communication between frontend and backend modules
MySQL Transactions	Atomic QR verification and transaction state updates
PowerShell	API and backend testing
E2E Test Scripts	Automated validation of QR and trust workflows

These technologies work together to implement the secure handover verification and peer-based trust functionality of ResourceHub.

3.3 Solution Overview

Module 4 consists of two primary components: QR Handover Verification and the Trust System.

QR Handover Verification

When an exchange request is accepted, the associated resource changes from AVAILABLE to RESERVED. The resource owner can then generate a temporary QR verification token.

The token is stored in the database with:

- Transaction ID
- Verification token
- Verification status

- Generation timestamp
- Expiration timestamp
- Verification timestamp
- Verifying user ID

During the physical handover, the requester scans the QR code.

The backend validates:

- Transaction existence.
- Token correctness.
- Requester identity.
- Token status.
- Token expiration.
- Transaction state.

If all conditions are satisfied:

QR Status = VERIFIED

The transaction can then proceed to completion.

The QR token cannot be reused after successful verification.

Trust System

After an exchange is completed, the participants can submit a review.

The review contains:

- Transaction ID.
- Reviewer ID.
- Reviewed user ID.
- Rating between 1 and 5.
- Optional review text.
- Creation timestamp.

The system prevents duplicate reviews from the same reviewer for the same transaction.

The resulting reviews are used to calculate the user's trust score and provide reputation information to other ResourceHub users.

Combined Module Workflow:

Exchange Request → ACCEPTED → RESERVED → QR Generated → QR Scanned → QR VERIFIED → COMPLETED → Review → Trust Score Update

3.4 Engineering Standards Applied

Several secure software engineering principles were applied during the development of Module 4.

Authentication and Authorization

Authenticated users are identified using JWT-based authentication. Backend authorization checks ensure that users can perform only operations appropriate to their role in a transaction.

Least Privilege

QR generation and verification operations are restricted according to transaction participation. Resource owners can generate QR codes, while requesters can verify them.

Defense in Depth

Multiple security controls are applied instead of depending on a single mechanism. These include authentication, authorization, secure token generation, expiration validation, transaction association, token status validation, and transaction protection.

Secure Randomness

QR verification tokens are generated using a secure random mechanism, making them difficult to predict.

Input Validation

The backend validates QR tokens, transaction identifiers, user relationships, review ratings, and review eligibility before modifying the database.

Database Transaction Integrity

Critical QR verification operations use database transactions and row locking to maintain consistent transaction states.

Duplicate Prevention

Database constraints and backend validation prevent duplicate reviews for the same transaction and reviewer.

Backend Enforcement

Security and business rules are enforced on the backend rather than relying solely on frontend controls.

Auditability and Traceability

Verification and review records contain transaction and user information that allows important activities to be associated with specific exchange operations.

3.5 Solution Justification

QR codes were selected as the handover mechanism because they provide a convenient way for two students to establish a digital confirmation during a physical exchange. The owner can display the QR code while the requester scans it using a mobile device.

A cryptographically secure random token was selected instead of using a predictable transaction identifier. This reduces the possibility of unauthorized users guessing valid verification values.

Expiration was implemented to limit the lifetime of QR verification codes. This ensures that a code generated for a particular handover does not remain valid indefinitely.

Authorization checks were implemented on the backend because frontend restrictions alone cannot reliably prevent malicious API requests. The backend therefore verifies the relationship between the authenticated user and the transaction.

QR verification was integrated directly into the transaction completion process because verification is meaningful only when it affects the actual transaction lifecycle. An accepted transaction cannot be completed until its handover has been verified.

The trust system was implemented using peer reviews because actual exchange participants are in a suitable position to provide feedback about the exchange experience. Duplicate-review prevention and rating validation improve the reliability of the collected reputation information.

Database transactions and row locking were used for critical state changes to reduce the possibility of race conditions and inconsistent transaction states.

Overall, the combination of QR verification and trust scoring provides both transaction-level assurance and user-level reputation, making ResourceHub more reliable than a simple resource listing platform.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Screenshots

The following screenshots can be included in the final report.

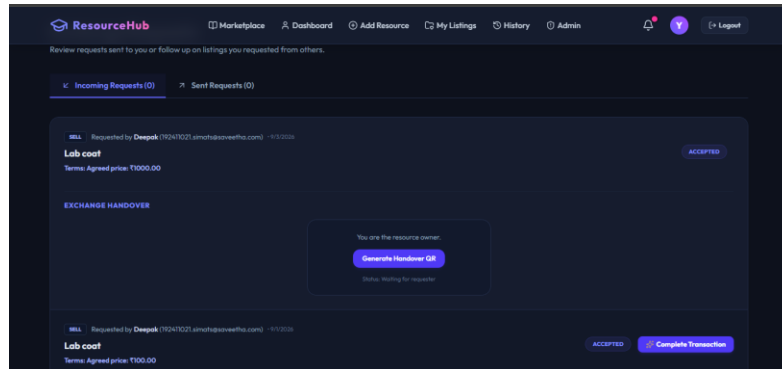


Figure 4.1: Accepted Exchange Request and QR Generation Interface

This screenshot demonstrates the owner-side QR handover interface after an exchange request has been accepted and the resource has been reserved. The interface provides the option to generate the temporary handover QR code.

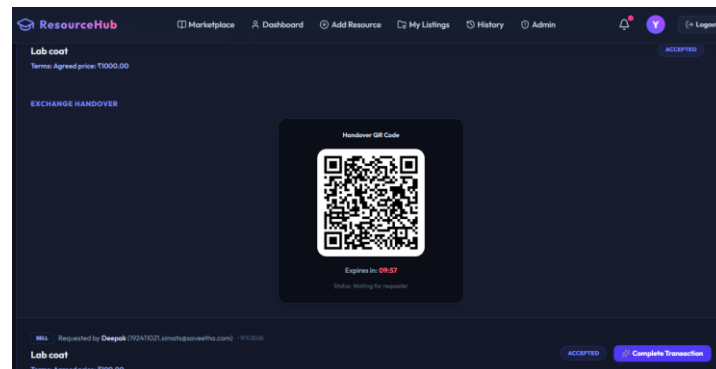


Figure 4.2: Generated Handover QR Code and Expiry Countdown

This screenshot demonstrates the generated QR verification code displayed to the resource owner together with the remaining validity period.

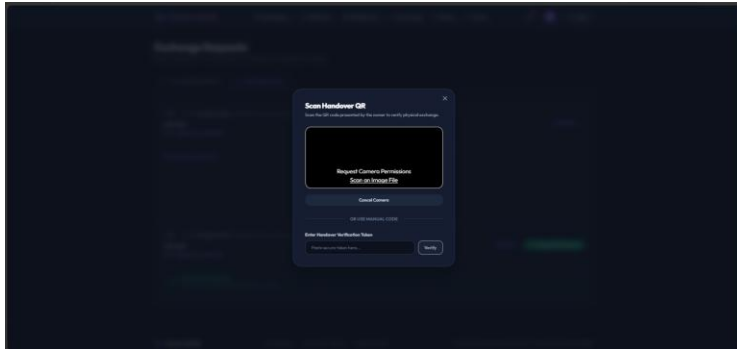


Figure 4.3: QR Scanning Interface

This screenshot demonstrates the requester-side QR scanning interface. The requester can use the device camera to scan the owner's handover QR code.

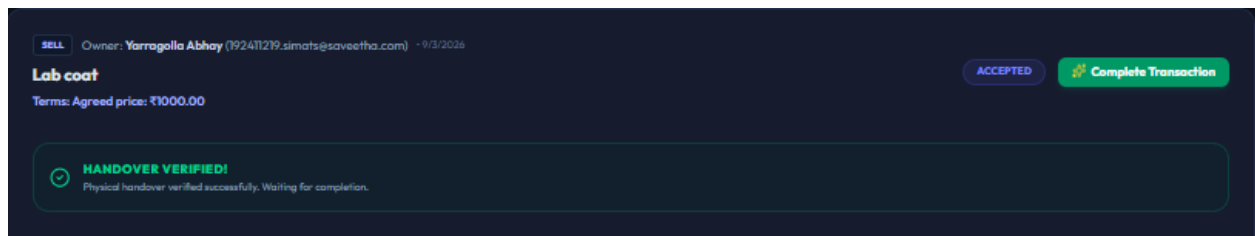


Figure 4.4: Successful QR Verification and Transaction Completion

This screenshot demonstrates the successful verification of the QR handover and the subsequent completion of the exchange transaction.

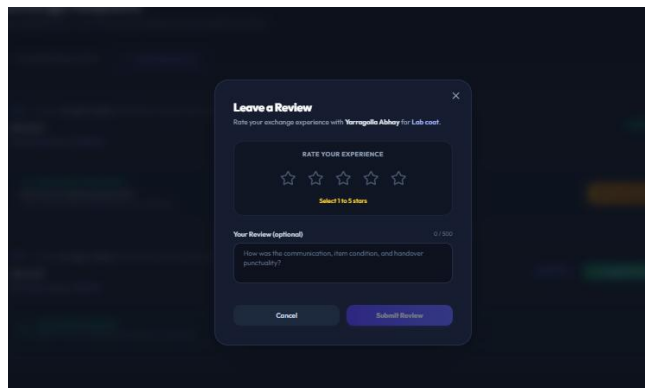


Figure 4.5: Review and Rating Interface

This screenshot demonstrates the review modal containing the one-to-five-star rating selector, optional review text, validation, and submission functionality.

4.2 Evaluation of Results

The implementation of Module 4 was evaluated through functional, security-oriented, database, and end-to-end test cases.

Test Case	Expected Result
Generate QR by resource owner	QR token is generated successfully
Generate QR by requester	Request is rejected with HTTP 403
Generate QR for invalid transaction	Request is rejected
Requester receives QR status	Secret verification token is not exposed
Complete transaction before QR verification	Completion is rejected with HTTP 400
Owner attempts to scan own QR	Verification is rejected
Unrelated student scans QR	Verification is rejected with HTTP 403
Complete verified transaction	Transaction completes successfully
Submit valid 5-star review	Review is stored successfully
Submit valid 4-star review	Review is stored and trust score updates
Duplicate review	Request is rejected with HTTP 409
Invalid rating	Request is rejected
Review database failure	Transaction rollback preserves database consistency
Review persistence after refresh	Submitted review remains displayed
QR regression test	All QR E2E checks pass
Frontend build	Vite compilation succeeds

The evaluation demonstrates that Module 4 successfully provides controlled physical handover verification and a functioning peer-based trust mechanism.

The QR component successfully prevented unauthorized generation and verification attempts, blocked transaction completion before handover verification, rejected expired tokens, and prevented token reuse.

The trust component successfully accepted valid ratings, prevented duplicate reviews, rejected invalid ratings, updated trust scores, and maintained review state after page refresh.

The automated validation also confirmed that QR functionality remained operational after the review and trust-system integration.

4.3 Challenges Faced

One challenge during implementation was ensuring that the QR verification process was correctly connected to the exchange transaction lifecycle. QR generation needed to be allowed only after a request had been accepted and the resource had been reserved.

Another challenge was implementing correct authorization for the two different QR operations. The resource owner must generate the QR code, while the requester must scan it. During frontend testing, an initial implementation displayed the handover QR option to both participants. This was corrected by separating the owner and requester views and enforcing the same authorization rules on the backend.

Handling token expiration was another challenge. The system needed to calculate the expiration time correctly and ensure that expired tokens could not be verified. Expired records were also updated to the EXPIRED state.

Preventing QR token reuse required additional status validation. A token that had already reached the VERIFIED state needed to be rejected if another verification attempt was made.

Integrating QR verification with transaction completion also required careful database handling. The system needed to ensure that the transaction could not be completed before successful QR verification while maintaining consistent resource and request states.

The trust system introduced additional challenges related to duplicate reviews, rating validation, trust-score updates, and database rollback. These issues required both backend validation and frontend state management.

Another challenge was ensuring that the frontend correctly reflected whether a completed transaction had already been reviewed. This was addressed by querying transaction reviews when completed requests are loaded, allowing the Review Submitted state to persist after page refresh.

These challenges improved practical understanding of authorization, database transactions, state management, frontend-backend integration, and security-oriented testing.

4.4 Proposed Improvements

Although the current implementation provides the required QR verification and trust functionality, several improvements could further strengthen ResourceHub.

First, the QR verification interface could provide stronger real-time feedback during scanning, including clearer camera permission messages and improved handling of poor lighting or camera failures.

Second, rate limiting could be introduced for QR verification endpoints to reduce automated attempts to guess or repeatedly submit verification tokens.

Third, the system could provide more detailed transaction notifications when QR verification succeeds, expires, or fails.

The trust system could also be extended to distinguish between different types of completed exchanges and provide more detailed reputation statistics.

Additional moderation controls could be introduced for reviews, allowing administrators to investigate inappropriate or abusive review content.

The platform could also provide stronger visual indicators for high-trust and low-trust users while ensuring that trust scores are presented fairly and transparently.

4.5 Future Recommendations

Future versions of ResourceHub could introduce more advanced QR handover mechanisms, including dynamically refreshed QR codes and stronger anti-replay protection.

The system could also support push notifications when an exchange is accepted, when a QR code is generated, when handover verification succeeds, and when a review is received.

The trust system could be expanded into a more comprehensive reputation model that considers completed exchange count, successful handovers, review consistency, cancellation history, and other non-sensitive transaction indicators.

Advanced review moderation could be introduced using reporting and administrative review mechanisms.

The system could also provide analytics dashboards showing exchange success rates, trust-score distributions, and verification activity.

Finally, the QR handover mechanism could be extended to support institution-specific verification requirements or integration with campus identity systems.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Working on Module 4 provided practical knowledge of implementing security mechanisms within a real-world web application. I learned how temporary verification tokens can be generated, associated with transactions, and validated securely.

The development of QR handover verification improved my understanding of authentication and authorization. I learned that identifying a logged-in user is different from determining whether that user is authorized to perform a particular transaction operation.

Another important learning outcome was understanding database transactions and row locking. QR verification and exchange completion involve multiple related state changes, and these operations must be handled carefully to maintain database consistency.

The review and trust system provided practical experience in implementing a reputation mechanism. I learned how ratings can be validated, stored, retrieved, and used to update a user's trust score.

The module also improved my understanding of frontend state management. Implementing QR generation, scanning, review submission, loading states, error handling, and persistent review status required coordination between React components and backend APIs.

The testing process improved my ability to design security-oriented test cases and perform end-to-end validation.

5.2 Challenges Encountered and Overcome

During the development of Module 4, several technical challenges were encountered.

One challenge was implementing the correct ownership and requester authorization rules for QR verification. Testing revealed that the frontend could display the wrong handover option to users. This was resolved by separating the owner and requester interfaces and reinforcing the authorization rules at the backend.

Another challenge involved QR token expiration and reuse protection. Testing different token states helped verify that generated, verified, and expired tokens were handled correctly.

Database transaction handling was another important challenge. Verification and transaction completion required multiple database operations to occur safely. Implementing transaction blocks and row locking improved consistency and reduced the risk of race conditions.

The review system introduced challenges involving duplicate submissions and trust-score calculations. These were addressed through database constraints, backend validation, and frontend submission-state management.

Frontend-backend integration also required debugging API responses, authentication headers, loading states, and error messages.

These challenges improved my problem-solving skills and provided practical experience in developing secure transaction-oriented web applications.

5.3 Applications of Engineering Standards

Security engineering principles were applied throughout the development of Module 4.

The principle of least privilege was applied by restricting QR generation to resource owners and QR verification to the corresponding requesters.

Defense in depth was achieved by combining JWT authentication, authorization checks, secure random tokens, token expiration, transaction association, token-status validation, and database transaction protection.

Backend validation was emphasized because frontend controls alone cannot provide reliable protection against malicious API requests.

Database integrity principles were applied through foreign-key relationships, unique constraints, transaction processing, and row locking.

Input validation was applied to QR verification requests and review ratings.

Duplicate prevention was implemented for reviews so that the same reviewer cannot submit multiple reviews for the same transaction.

The system also follows the principle of separation of concerns by maintaining separate controllers, routes, frontend services, and UI components for exchange, QR verification, and reviews.

5.4 Industry Insights

The development of Module 4 provided an understanding of how security and trust mechanisms are implemented in practical transaction-based applications.

One important industry insight is that authentication and authorization are different security requirements. A user being logged in does not automatically mean that the user should be permitted to perform every operation.

Another important lesson is that security controls must be enforced on the server. Frontend interfaces can improve usability, but backend validation is necessary because clients can be modified or bypassed.

The QR verification mechanism demonstrated how temporary tokens can connect a physical event with a digital transaction. This concept can be applied to delivery confirmation, event entry, equipment handover, and other real-world workflows.

The trust system demonstrated the importance of reputation in peer-to-peer platforms. Users often need information about another participant before committing to an exchange.

The project also demonstrated the importance of database consistency. In transaction-oriented applications, multiple related changes must be performed atomically to prevent inconsistent states.

5.5 Conclusion of Personal Development

Working on Module 4 significantly improved my understanding of secure web application development and transaction management.

The module provided practical experience in QR verification, secure token generation, authorization, token expiration, transaction protection, database transactions, peer reviews, rating validation, and trust-score management.

The development process also improved my skills in React frontend development, REST API integration, MySQL database handling, debugging, automated testing, and end-to-end validation.

Overall, the experience helped me understand how security, usability, database consistency, and user trust can be combined to create a reliable real-world software system.

CHAPTER 6

CONCLUSION

Module 4, QR Verification & Trust System, provides an important security and reputation layer within the ResourceHub platform.

The QR verification component creates a controlled connection between an accepted exchange transaction and its physical handover. A temporary verification token is generated by the authorized resource owner and presented as a QR code. The requester scans the code during the handover, after which the backend validates the token, transaction association, requester identity, expiration status, and verification state.

The system prevents unauthorized QR generation, self-verification, unrelated-user verification, expired-token usage, and token reuse. QR verification is also enforced before transaction completion, ensuring that an accepted exchange cannot be incorrectly marked as completed without the required handover verification.

The trust component complements QR verification by allowing participants to rate each other after completed exchanges. The system supports one-to-five-star ratings and optional written reviews while preventing duplicate and invalid submissions. Trust scores are updated using the implemented rating mechanism and displayed as part of the user's reputation.

The implementation was validated through comprehensive end-to-end testing. QR generation, authorization, expiry, verification, transaction completion, duplicate prevention, review submission, trust-score updates, rollback handling, review persistence, and frontend compilation were successfully tested.

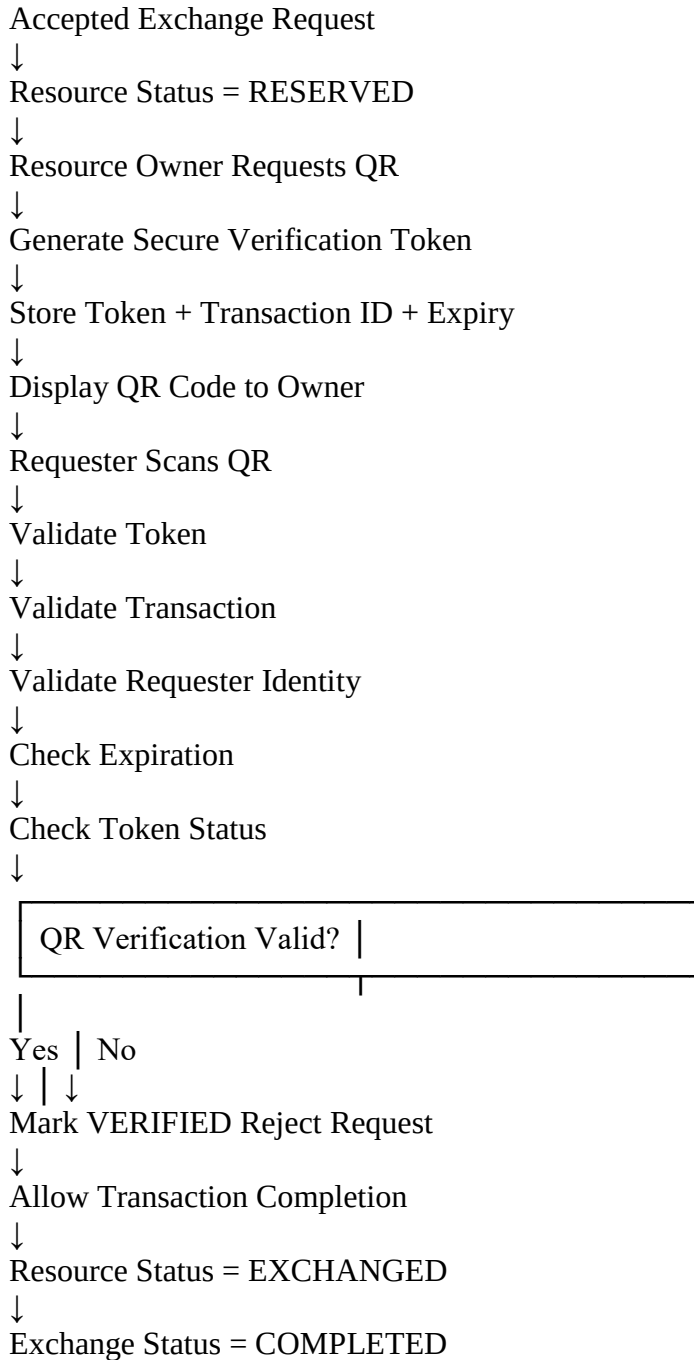
Overall, Module 4 improves the reliability and accountability of ResourceHub by addressing both transaction-level verification and user-level trust. It transforms ResourceHub from a basic academic resource exchange platform into a more secure and trustworthy peer-to-peer exchange system.

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12. Technical documentation for QR-code generation and QR-code scanning libraries used during ResourceHub implementation.
13. ResourceHub project database schema and API implementation documentation.
14. ResourceHub Module 4 E2E validation and testing documentation.

APPENDICES

Appendix A — QR Handover Verification Workflow



Appendix B — Trust System Workflow

Completed Exchange

↓

Check Review Eligibility
↓
Identify Reviewer
↓
Identify Reviewed User
↓
Select Rating (1–5)
↓
Enter Optional Review
↓
Validate Rating
↓
Check Duplicate Review
↓
Store Review
↓
Calculate Updated Trust Score
↓
Update User Trust Score
↓
Return Updated Reputation
↓
Display Review Submitted

Appendix C — QR Verification Implementation

The QR verification implementation contains the following major operations:

1. Generate a secure verification token.
2. Associate the token with an exchange transaction.
3. Set the token expiration time.
4. Restrict generation to the resource owner.
5. Validate the requester during scanning.
6. Reject self-scanning by the resource owner.
7. Reject unrelated users.
8. Reject expired tokens.
9. Prevent reuse of verified tokens.
10. Mark successful verification as VERIFIED.
11. Record the verification timestamp and verifying user.
12. Use database transaction protection during state updates.

Appendix D — Trust Score and Review Implementation

The review implementation contains the following major operations:

1. Validate that the exchange has been completed.
2. Identify the authenticated reviewer.
3. Identify the participant being reviewed.
4. Validate the rating range from 1 to 5.
5. Accept optional review text.
6. Prevent duplicate reviews for the same transaction and reviewer.
7. Insert the review record.
8. Calculate the updated review statistics.
9. Update the reviewed user's trust score.
10. Return the updated trust information to the frontend.

The frontend review interface provides star selection, hover feedback, review text input, character counting, submission loading state, error handling, and persistent review status after page refresh.

Appendix E — API Endpoints Related to Module 4

API Method	Purpose
POST	Generate QR handover verification token
POST	Verify QR handover token
GET	Retrieve QR verification status
POST	Complete verified exchange transaction
POST	Create peer review
GET	Retrieve user reviews
GET	Retrieve transaction reviews

The exact endpoint paths should be copied from the implemented ResourceHub route files before final report submission.

Appendix F — Database Structures Related to Module 4

The QR verification functionality uses the `qr_verifications` table to maintain:

- Transaction ID
- Verification token
- Verification status
- Generated timestamp
- Verified timestamp
- Verifying user ID
- Expiration timestamp

The trust and review functionality uses the `reviews` table to maintain:

- Transaction ID

- Reviewer ID
- Reviewed user ID
- Rating
- Review text
- Creation timestamp

The users table maintains the calculated `trust_score` associated with each student.

The exchange transaction state is maintained through the `exchange_requests` and `resources` tables.

Appendix G — Testing Checklist

Test	Expected Outcome
Owner generates QR	QR generated successfully
Requester generates QR	Access denied
Invalid transaction QR	Request rejected
QR status request	Secret token protected
Complete before QR verification	Completion rejected
Owner scans own QR	Scan rejected
Unrelated student scans QR	Access denied
Correct requester scans QR	Verification succeeds
Reuse verified QR	Request rejected
Expired QR	Request rejected and token marked EXPIRED
Verified transaction completion	Exchange completed
Submit 5-star review	Review stored
Submit 4-star review	Trust score updated
Submit duplicate review	HTTP 409 rejection
Submit invalid rating	Validation failure
Review database failure	Transaction rollback
Refresh completed request	Review state persists
M7 QR regression	All tests pass
Client build	Vite build succeeds

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.2–4.7)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.1–3.3)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)